

## Fisetin inhibits various attributes of angiogenesis in vitro and in vivo--implications for angioprevention.

. . . . .

Studies have shown that fisetin, a small phytochemical molecule, has antitumor activity; however, its antiangiogenic activity has not yet been examined. Accordingly, herein, we investigated the antiangiogenic efficacy and associated mechanisms of fisetin in human umbilical vein endothelial cells (HUVECs). Fisetin (10-50  $\mu$ M) strongly inhibited the regular serum plus growth supplement- and vascular endothelial growth factor (VEGF)-induced growth (up to 92%,  $P < 0.001$ ) and survival (up to 16%,  $P < 0.001$ ) of HUVEC in a dose- and time-dependent manner. Fisetin also caused cell cycle arrest at G(1) (strong) and G(2)/M (moderate) phases together with a decrease in cyclin D1 and an increase in p53 levels. Fisetin-caused cell death was accompanied by decreased expression of survivin and an increase in cleaved levels of caspases-3 and -7 and poly-(ADP-ribose) polymerase along with an increased ratio of Bax to Bcl-2. Furthermore, fisetin inhibited capillary-like tube formation on Matrigel (up to 85%,  $P < 0.001$ ) as well as migration (up to 66%,  $P < 0.001$ ), which were associated with decreased expression of endothelial nitric oxide synthase (eNOS) and VEGF in HUVEC. It also decreased the expression of eNOS, VEGF, inducible nitric oxide synthase, matrix metalloproteinase-2 and -9 in A549 and DU145 human cancer cells. In vivo matrigel plug assay in mice showed significant decrease in size (up to 43%,  $P < 0.001$ ), vascularization and hemoglobin content (up to 94%,  $P < 0.001$ ) in the plugs from fisetin-treated, compared with control mice. Overall, these results suggest that fisetin inhibits various attributes of angiogenesis, which might contribute to its reported antitumor effects, and therefore, fisetin warrants further investigation for its angiopreventive potential toward cancer control.